Chest Pain / Acute Coronary Syndrome

- **Full Vital Sign Assessment**
- **12-Lead ECG**
- Consider Right Sided and Posterior ECG

Consider Oxygen Therapy
Titrate to maintain Sp02 of **90%**

EMR

ACS Inclusion Criteria:

- Ischemic-Type Symptoms/Pain
- Identified ACS on 12-lead:
  - STEMI / NSTEMI / Ischemic changes (e.g., T wave inversions, ST depressions, new LBBB)
- Non-specific presentation and cannot exclude cardiac origin
- Epigastric pain, syncope, palpitations with possible cardiac cause AND no clear alternative
- Hymodynamic compromise or arrhythmias (with or without chest pain) if suspicion of cardiac origin is high

Aspirin
162-324mg PO

EMR

Consider Tele EMS

If SBP >100mmHg
Consider GTN
0.4mg SL
Repeat every 3-5 mins
(max 1.2mg)
(reassess after each dose)

EMT

Consider Morphine
For severe pain only
1-2mg IV increments

EP

If STEMI on 12-Lead

- Apply Pads
- Expedite patient transfer

GTN Contraindicated:

- Phosphodiesterase Inhibitor use (typically used for erectile dysfunction)
- Inferior STEMI with suspected Right Ventricular involvement

SBP < 90mmHg?

Consider IV Fluid Bolus:
250ml IV/IO bolus
Monitor closely for pulmonary edema

Patient Care Goals

1. Conduct a full vital signs assessment and 12-Lead ECG to quickly identify ST-elevation myocardial infarction (STEMI) and/or Occlusion Myocardial Infarction (OMI).
2. Determine the time of symptom onset.
3. Expedite patient transport to an appropriate cardiac hospital.

Patient Inclusion & Exclusion Criteria

Inclusion Criteria

- Patients presenting with **ischemic-type symptoms**, with or without ECG changes suggestive of ischemia.
  - Ischemic-type symptoms include, but not limited to:
    - Chest pain / pressure or discomfort
    - Radiation to arms, jaw, back, shoulders
    - Dyspnea / shortness of breath
    - Diaphoresis / sweating
    - Nausea, vomiting
    - Dizziness, syncope, palpitations
- Identified **ACS on 12-lead ECG**:
  - STEMI
  - Non-STEMI / ischemic ECG changes (e.g., T wave inversions, ST depressions, new LBBB)
- Patients with **non-specific chest pain** or **atypical presentation** when cardiac origin cannot be excluded.
- Presentations **of epigastric discomfort**, **syncope**, or **palpitations** when a cardiac cause is among differential, and **no clear alternative non-cardiac cause is evident** at first evaluation.
- Patients with **signs of hemodynamic compromise** or arrhythmias, even if chest pain is absent**, if suspicion of an acute cardiac etiology is high.**

Exclusion Criteria

- Patients with a clear non-cardiac cause of chest pain.

Key Patient Assessment Points

1. **Conduct a Full Vital Signs Assessment**
   - Perform a 12-Lead ECG.
   - Consider right-sided and posterior ECG.
2. **Determine Time of Symptom Onset**
   - Determine the time of symptom onset as this will influence patient management.
3. **Evaluate and determine pain type**
   - Investigate chest pain using OPQRST assessment and associated signs and symptoms.
   - Determine between ischemic, non-ischemic and non-cardiac pain.
4. **Determine Triage Category**
   - Determine Triage Category as per Patient Triage Categories CPG.
     - Red Priority:
       - Signs of Severe Shock
       - ST elevation myocardial infarction (STEMI)
       - Chest Pain or Dysrhythmias or other ECG changes WITH significant cardiovascular compromise.

Patient Management and Interventions

1. **Consider Oxygen Therapy**
   - Target SpO2 of 90%.
   - Administer oxygen using an appropriate oxygen delivery mask.
2. **Consider ALS back-up**
   - Consider the requirement for ALS back-up as per Universal Care CPG.
   - ALS back-up should be called if:
     - The patient is hemodynamically unstable. Or/

- The patient is in VT, or is having 'runs' of VT. Or/
- The patient has severe difficulty of breathing. Or/
- The patient is agitated to a level which inhibits patient care. Or/
- The EMS team suspect the patient is likely to deteriorate into cardiac arrest.

3. **ACS General Management**
   - If patient presents with ischemic type chest pain (see table below) or meets the ACS inclusion Criteria
   - **Consider Aspirin**
     - Administer Aspirin 162-324mg PO.
     - The standard aspirin dose is 324mg (4 tablets)
     - If the patient has taken aspirin within the last 60 minutes:
       - Confirm the dose they have received.
       - Administer additional Aspirin to reach a maximum total dose of 324mg.
   - **Consider GTN**
     - If ischemic chest pain is present, administer GTN.
     - GTN 0.4mg SL, repeat every 3-5 mins while symptoms persist and if systolic BP remains ≥100 mmHg. (max 1.2mg)
     - Reassess vital signs before repeating dose.
       - Note Contraindications: Phosphodiesterase inhibitor use (typically used for erectile dysfunction), inferior STEMI with suspected right ventricular involvement.
   - **Consider Morphine**
     - Indicated for severe pain only.
     - Administer Morphine IV/IO 1-2mg increments, repeat every 5-10 minutes, maximum 10mg.
   - **If STEMI on 12-Lead ECG:**
     - Apply AED Pads
       - In preparation for cardiac arrest
     - Continuous ECG Monitoring
       - Routinely use 4-lead ECG monitoring as well as applying the pads so that pacing and electrical cardioversion can be easily performed.
     - Expedite transport
       - Expedite transport and prioritize transfer of the patient to hospital—however, do not neglect proper patient assessment and management.
       - Coordinate with Dispatch 2 to ensure that the patient is transferred to an appropriate cardiac hospital.
   - **If Non-STEMI on 12-Lead ECG:**
     - Perform serial 12-leads (take a 12-lead with each set of vitals) to observe for progression.
     - Identification of Occlusion Myocardial Infarction (OMI):
       - OMI can indicate acute coronary occlusion requiring urgent intervention despite the absence of classic STEMI criteria—contact Telemedicine consultation for confirmation and escalation if required.
       - Recognize Occlusion Myocardial Infarction (OMI) that may not present as STEMI on ECG; look for subtle signs like hyperacute T-waves, reciprocal changes, or ST depression in multiple leads.
4. **If Patient Hemodynamically Unstable**
   - **Consider Normal Saline Fluid Bolus**
     - If sBP <90mmHg:
       - Consider Normal Saline 250ml IV/IO
       - Monitor closely for fluid overload/pulmonary edema.
       - Stop fluid bolus if pulmonary edema develops.
       - Fluid Bolus may be repeated as required.
5. **Contacting Telemedicine**
   - In 12-Lead ECGs where the diagnosis is non-specific or unclear or if the clinician is unsure, telemedicine should be consulted for confirmation or second opinion.

- Telemedicine should be used routinely by EMTs when the diagnosis is not clear STEMI.

6. **Patient Reassessment**
   - Frequently Reassess the patient as per Universal Care CPG.
   - Continuous ECG monitoring is required for all patients with cardiac chest pain.

Patient Safety Considerations

- Investigate uncommon indicators of cardiac chest pain, including pain radiating to arms or jaws, dizziness, sweating, vomiting, and nausea.
- Phosphodiesterase inhibitors (such as Viagra) may be used for other conditions—closely investigate the patient's medications.
- GTN should be avoided in tachy and brady arrhythmias; management of the rate should be prioritized.
- Administer fluids cautiously; small amounts of fluid in patients with cardiac failure may result in fluid overload.

Clinical Insights & Additional Information

- Common features of ischemic/non-ischemic and non-cardiac type chest pain:[11-18]

	Ischemic (e.g., Angina, ACS, STEMI, n-STEMI)	**Non-Ischemic** (e.g., Pericarditis, Aortic Dissection, Pulmonary Embolism)	**Non-Cardiac** (e.g., Gastrointestinal, Musculoskeletal, Respiratory)
Location	Substernal or retrosternal, may radiate to arms, neck, jaw, shoulders, or back	Variable; retrosternal (pericarditis), chest/back (aortic dissection), lateral (PE)	Variable; retrosternal (GERD), localized (musculoskeletal), lateral (respiratory)
Quality	Pressure, squeezing, heaviness, tightness, burning	Sharp, pleuritic (pericarditis, PE); tearing/ripping (aortic dissection)	Burning (GERD); sharp, aching (musculoskeletal); sharp, pleuritic (respiratory);
Duration	2-30 minutes	Hours to days (pericarditis); sudden and persistent (aortic dissection, PE)	Minutes to hours (GERD); variable (musculoskeletal)
Onset	Gradual or sudden	Sudden (aortic dissection, PE); gradual (pericarditis)	Gradual or sudden
Worsening Factors	Exertion, emotional stress	Inspiration, supine position (pericarditis); deep breath (PE)	Meals, lying down (GERD); movement, palpation (musculoskeletal); inspiration (respiratory);
Relieving Factors	Rest, nitroglycerin	Leaning forward (pericarditis)	Antacids (GERD); NSAIDs (musculoskeletal)
Associated Symptoms	Dyspnea, nausea, vomiting, diaphoresis, palpitations	Fever, pericardial rub (pericarditis); neurologic symptoms, unequal blood pressure/pulses (aortic dissection); dyspnea, hemoptysis, tachycardia (PE)	Regurgitation (GERD); tenderness (musculoskeletal); cough, fever (respiratory)

- Transport Decision:
  - Transport the patient to the closest most appropriate cardiac hospital with ongoing care, monitoring vital signs with attached SpO2 and ECG, and ensuring timely transfer.
- Special Considerations:

- STEMI patients are at high risk of cardiac arrest and arrhythmias; apply pads for monitoring and potential pacing/cardioversion.
- Non-STEMI patients may require serial 12-lead ECGs to observe progression; contact Telemedicine for OMI confirmation.
- Avoid GTN in tachy or brady arrhythmias; prioritize rate management.

Documentation

- Identification: Full patient name, age, and gender.
- Mechanism / Medical Complaint: Presenting symptoms (e.g., chest pain, radiating pain), initial vital signs.
- Injuries / Information: Relevant history (e.g., cardiac risk factors, medications).
- Signs:
  - Full vital signs including pulse, respiratory rate, SpO2, sBP.
  - ECG findings (e.g., STEMI, OMI indicators).
- Treatment:
  - Oxygen administration, Aspirin, GTN, Morphine, and NaCl doses.
  - All interventions: Pads application, ECG monitoring, and response.
  - Justification if interventions were withheld.
- Allergies: Documented allergies and drug interactions.
- Medications: All pre-existing medications and any administered in the field.
- Background: Relevant medical history, comorbidities, and recent procedures.
- Other:
  - Communication with Telemedicine (time and instructions).
  - Response to interventions.
  - Consideration of reversible causes and actions taken.

Key Performance Indicators

1. All patients with cardiac chest pain should receive a 12-lead ECG.
2. Administer aspirin to all ACS patients (where appropriate).

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